

Flexible Cable Simulation in NVIDIA Isaac Sim

Rigid Capsule-Chain vs Deformable Body for Dual-Arm Manipulation

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Outline

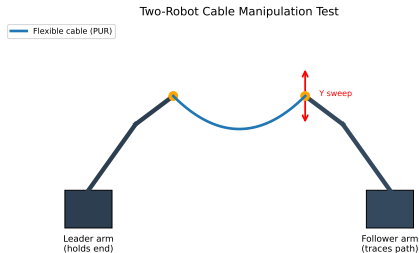
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Problem Statement

- Isaac Sim **lacks native 1D deformable** support
- Need: realistic cable sim for **robotic manipulation**
- Two candidate models to evaluate:
 - ① **Rigid capsule-chain** (material-driven joints)
 - ② **FEM deformable body** (native soft-body)

Material: PUR (polyurethane) — the most common jacket for robot / drag-chain cables ($E \approx 30$ MPa).

Goal: pick the best model for RL-based cable manipulation.

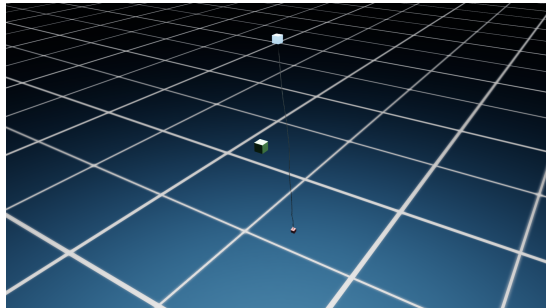


Method 1: Rigid Capsule-Chain Model

Architecture:

- Chain of N rigid capsules + D6 joints
- Bending: $K = EI/L_{\text{seg}}$ (beam theory)
- Axial MSD springs: $k_s = EA/L_{\text{seg}}$
- Twist DOF free (Govoni)

Validated: reproduces Govoni Table 1
stays stable where their explicit solver diverges
(800× larger timestep).



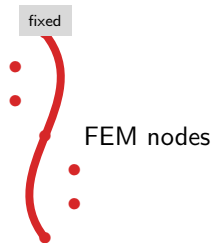
Hanging cable, 200 links, $E = 30$ MPa

Method 2: FEM Deformable Body

Approach:

- Single deformable cylinder mesh
- FEM tetrahedral simulation
- Native self-collision + axial stretch
- Same PUR material (E , ν , density)
- GPU-accelerated

Setup: `DeformableObjectCfg` +
`MeshCylinderCfg` +
`DeformableBodyMaterialCfg`



Continuous mesh, not discrete links

Two-Robot Cable Manipulation Test

Setup:

- Two Franka Panda arms facing each other
- Each grasps one cable end
- Leader holds still
- Follower traces 0.15 m sweep @ 0.5 Hz

[figure pending]
figures/motion_transmission.pdf
run run_all.sh

We measure:

- Motion transmission (command vs response)
- Span error (inextensibility)
- Reaction force at follower

Comparison Criteria

Criterion	Capsule-chain	Deformable body
Stability ($\Delta t = 1/240$ s)	Validated	To test
Inextensibility	Exact (rigid)	Stretches (FEM)
Axial elasticity	No	Yes
Self-collision	Approximate	Native
Visual realism	Good	Excellent
Compute cost	Low-moderate	High
RL scalability	High (many envs)	Low

Decision rule

Capsule-chain for large-scale RL training (cheap, inextensible).

Deformable for tasks needing knotting, wrapping, or force feedback.

Comparison: Quantitative

[figure pending]

figures/span_error_compare.pdf

run run_all.sh

Inextensibility (span error)

[figure pending]

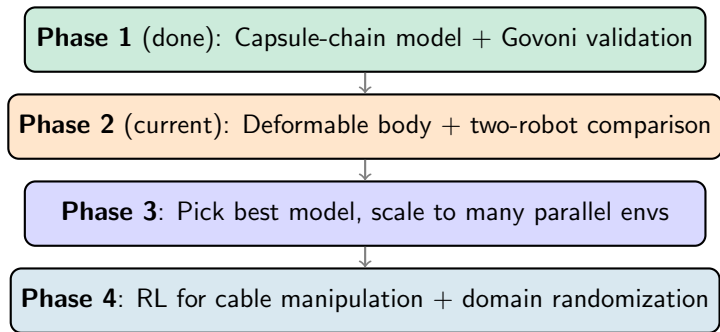
figures/compute_cost_compare.pdf

run run_all.sh

Force transmitted to follower arm

All values measured from the two-robot Isaac Sim runs.

Research Roadmap



RL background: DDPG, TD3, SAC, PPO

Two published papers (ISA Transactions, Eng. Applications of AI)

Summary

- ① **Two cable models** built on common PUR material ($E = 30 \text{ MPa}$)
- ② **Capsule-chain**: validated, cheap, inextensible — RL-ready
- ③ **Deformable body**: physical stretch + self-collision — higher fidelity
- ④ **Two-robot test** + criteria to choose between them objectively
- ⑤ **Next**: measured comparison $\rightarrow$ scale up $\rightarrow$ RL manipulation

Thank you. Questions?

-  A. Govoni et al., “Performance Analysis of a MSD DLO Model in Robotic Simulation Frameworks,” *arXiv:2504.13659*, 2025.
-  M. Bergou et al., “Discrete Elastic Rods,” *ACM TOG (SIGGRAPH)*, 27(3), 2008.